

BAULKHAM HILLS HIGH SCHOOL

YEAR 11

HSC ASSESSMENT TASK

December 2008

MATHEMATICS
EXTENSION 1

Time allowed - 70 minutes

DIRECTIONS TO CANDIDATES

- Attempt ALL questions.
- Allocated marks are indicated for each question.
- All necessary working should be shown.
- Write your teacher's name and your name on the cover sheet provided.
- At the end of the exam, staple your answers in order behind the cover sheet provided.

Question 1

- a) Solve $\cos 2x = \sin x$ for $0 \leq x \leq 2\pi$ 4
- b) (i) Find $\int \frac{1}{4(2x+1)^3} dx$ 2
- (ii) Find $\int \sin^2 4x dx$ 2

Question 2

Differentiate $x \sin x$. Hence or otherwise find $\int x \cos x dx$ 3

Question 3

Find the Cartesian equation of the curve with parametric equations:

- a) $x = t^2 + 2$, $y = 3t$ 1
- b) $x = \cos t$, $y = \sin 2t$ 2

Question 4

If α , β and γ are the roots of $2x^3 - 5x^2 + 3x - 15 = 0$, find the value of

- a) $\alpha + \beta + \gamma$ 1
- b) $\alpha\beta\gamma$ 1
- c) $(\alpha - 1)(\beta - 1)(\gamma - 1)$ 2

Question 5

Find $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{x^2}$, showing all necessary working 2

Question 6

If $P(x) = x^3 - 3x + 2$

- a) Factorise $P(x)$ completely 3
- b) Solve $x^3 - 3x + 2 > 0$ 2

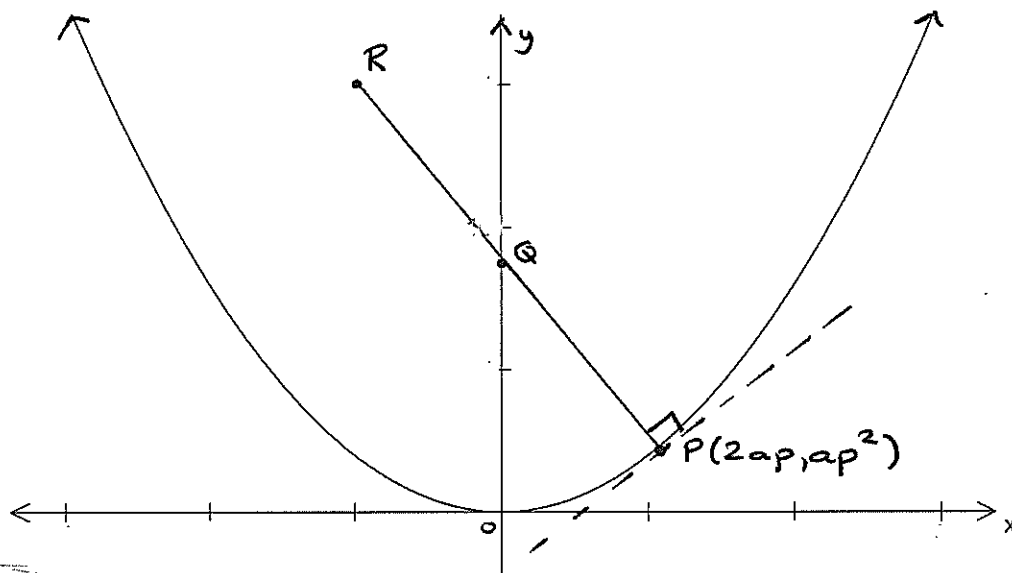
Question 7

Given that $P(x) = x^3 + ax^2 + bx + c$ and

- * $P(0) = 1$
- * $x - 1$ is a factor of $P(x) + 1$
- * $x + 1$ is a factor of $P(x) - 1$
- Determine the values for a , b and c 4

Question 8

The point $P(2ap, ap^2)$ is a variable point on the parabola $x^2 = 4ay$. The normal at P cuts the y -axis at Q . PQ is then produced to R such that $PQ = QR$.



- Show that the equation of the normal at P is $x + py = ap^3 + 2ap$ 2
- Find the coordinates of Q and R 3
- Deduce that the locus of R is a parabola and state the coordinates of its vertex. 3

Question 9

The area between the curve $y = 1 + 2\cos x$, the x -axis and the lines $x = 0, x = \frac{\pi}{2}$ is rotated about the x -axis. Find the exact volume of the solid of revolution generated. 4

Question 10

The cubic $y = x^3 + ax^2 + bx + c$ has three distinct roots α, β and γ which form an arithmetic progression, with $\alpha < \beta < \gamma$.

Given that $\beta = k$, find the coordinates of the point of inflexion of the curve, in simplest form. 3

End of exam

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln(x + \sqrt{x^2 - a^2}), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln(x + \sqrt{x^2 + a^2})$$

NOTE: $\ln x = \log_e x$, $x > 0$

Q1.

a) $1 - 2\sin^2 x = \sin x$ 1

$$0 = 2\sin^2 x + \sin x - 1$$

$$0 = (2\sin x - 1)(\sin x + 1)$$
 1

$$\sin x = \frac{1}{2}$$

$$\sin x = -1$$

$$x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$$

b) i) $\frac{1}{4} \int (2x+1)^{-3} \cdot dx$

$$= \frac{1}{4} \cdot \frac{(2x+1)^{-2}}{-2 \times 2} + c$$

$$= -\frac{1}{16} \cdot (2x+1)^{-2} + c$$

or $\frac{-1}{16(2x+1)^2} + c$

2 (-1 each error). (ignore c)

ii) $\int \sin^2 4x \cdot dx$

$$\left[\sin^2 A = \frac{1}{2}(1 - \cos 2A) \right]$$

$$= \int \frac{1}{2}(1 - \cos 8x) \cdot dx$$
 1

$$= \frac{1}{2} \left(x - \frac{1}{8} \sin 8x \right) + c$$
 1

(ignore c)

Q2.

$$\frac{d}{dx} (x \sin x)$$

$$= x \cdot \cos x + \sin x \cdot 1$$

$$= x \cos x + \sin x$$
 1

$$\int x \cos x + \sin x \cdot dx = x \sin x + c$$
 1

$$\int x \cos x \cdot dx + \int \sin x \cdot dx = x \sin x + c$$

$$\int x \cos x \cdot dx = x \sin x - \int \sin x \cdot dx$$

$$= x \sin x + \cos x + c$$
 1

Q3.

a) $t = \frac{y}{3}$

$$x = \left(\frac{y}{3}\right)^2 + 2$$

$$x = \frac{y^2}{9} + 2$$

b) $x = \cos t$



$$y = 2 \sin t \cos t$$

$$\therefore y = 2 \cdot \sqrt{1-x^2} \cdot x \quad \text{or} \quad y = 2x\sqrt{1-x^2}$$

Alternative method :

$$\cos t = x$$

$$y = 2 \sin t \cos t \quad \therefore \sin t = \frac{y}{2x}$$

$$\sin^2 t + \cos^2 t = 1$$

$$\left(\frac{y}{2x}\right)^2 + x^2 = 1$$

$$\text{or} \quad \frac{y^2}{4x^2} + x^2 = 1$$

or equiv.

Q4.

a) $\alpha + \beta + \gamma = \frac{5}{2}$

b) $\alpha\beta\gamma = \frac{15}{2}$

c) $(\alpha-1)(\beta-1)(\gamma-1) = (\alpha-1)(\beta\gamma - \beta - \gamma + 1)$

$$= \alpha\beta\gamma - \alpha\beta - \alpha\gamma + \alpha - \beta\gamma + \beta + \gamma - 1$$

$$= \alpha\beta\gamma - (\alpha\beta + \beta\gamma + \alpha\gamma) + (\alpha + \beta + \gamma) - 1$$

$$= \frac{15}{2} - \left(\frac{3}{2}\right) + \frac{5}{2} - 1$$

$$= \frac{15}{2}$$

Q5.

$$\lim_{x \rightarrow 0} \frac{1 - (1 - 2\sin^2 x)}{x^2}$$

$$= \lim_{x \rightarrow 0} \frac{2 \sin^2 x}{x^2} \leftarrow 1$$

$$= \lim_{x \rightarrow 0} 2 \left(\frac{\sin x}{x} \right)^2$$

$$= 2 \times 1^2 = \underline{2} \leftarrow 1$$

(Answer only: 1 mark)

Q6.

a) $P(1) = 1^3 - 3 + 2 = 0$
 $\therefore x-1$ is a factor

$$\begin{array}{r} x^2 + x - 2 \\ x-1 \overline{) x^3 - 3x + 2} \\ \underline{x^3 - x^2} \\ x^2 - 3x \\ \underline{x^2 - x} \\ -2x + 2 \\ \underline{-2x + 2} \\ 0 \end{array}$$

One factor = 1

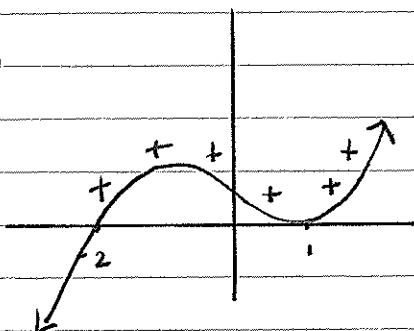
Long div / Quad factor = 1

Final line = 1

$$P(x) = (x-1)(x^2 + x - 2)$$

$$\begin{aligned} &= (x-1)(x+2)(x-1) \\ &= (x-1)^2(x+2) \end{aligned} \left. \vphantom{\begin{aligned} &= (x-1)(x+2)(x-1) \\ &= (x-1)^2(x+2) \end{aligned}} \right\} \text{accept either.}$$

b)



$$x > -2 \text{ but } x \neq 1$$

OR

$$-2 < x < 1, x > 1$$

2

Q7.

$$P(x) = x^3 + ax^2 + bx + c$$

$$P(0) = 1 : 0 + 0 + 0 + c = 1 \quad \underline{c=1}$$

$$P(x) = x^3 + ax^2 + bx + 1$$

$$P(x) + 1 = x^3 + ax^2 + bx + 2$$

$$P(1) + 1 = 1 + a + b + 2 = 0$$

$$a + b = -3 \quad \dots (1)$$

$$P(x) - 1 = x^3 + ax^2 + bx$$

One or both
simult eqns

$$P(-1) - 1 = -1 + a - b = 0$$

$$a - b = 1 \quad \dots (2)$$

Solving simult: $\underline{a = -1}, \underline{b = -2}$

Q8.

$$a) y = \frac{x^2}{4a}, \quad \frac{dy}{dx} = \frac{2x}{4a} = \frac{2(2ap)}{4a} \quad \text{at } P. \quad |$$

$= p$

$$\text{Normal : } m = -\frac{1}{p}$$

$$y - ap^2 = -\frac{1}{p} (x - 2ap) \quad |$$

$$py - ap^3 = -x + 2ap$$

$$x + py = ap^3 + 2ap.$$

b) At Q : $x=0$

$$0 + py = ap^3 + 2ap$$
$$y = ap^2 + 2a$$

$$\therefore Q(0, ap^2 + 2a)$$

no mark for x .

At R : $x = -2ap$

$$y = ap^2 + 2a + 2a = ap^2 + 4a$$

$$\therefore R(-2ap, ap^2 + 4a)$$

c) At R : $x = -2ap$, $p = \frac{-x}{2a}$

$$y = ap^2 + 4a$$

$$y = a\left(\frac{-x}{2a}\right)^2 + 4a$$

$$y = \frac{x^2}{4a} + 4a$$

$$\Leftrightarrow x^2 = 4a(y - 4a) \quad \left\{ \begin{array}{l} \text{a parabola.} \end{array} \right.$$

Vertex $(0, 4a)$.

Q9. $V = \pi \int_0^{\pi/2} (1 + 2 \cos x)^2 \cdot dx$

Now $(1 + 2 \cos x)^2 = 1 + 4 \cos x + 4 \cos^2 x$

$$= 1 + 4 \cos x + 4 \cdot \frac{1}{2} (1 + \cos 2x)$$

$$= 1 + 4 \cos x + 2 + 2 \cos 2x$$

$$= 3 + 4 \cos x + 2 \cos 2x$$

$$\therefore V = \pi \int_0^{\pi/2} 3 + 4 \cos x + 2 \cos 2x \cdot dx$$

$$= \pi \left[3x + 4 \sin x + \sin 2x \right]_0^{\pi/2}$$

$$= \pi \left(\frac{3\pi}{2} + 4 \underbrace{\sin \frac{\pi}{2}}_1 + \underbrace{\sin \pi}_0 - (0+0+0) \right)$$

$$= \pi \left(\frac{3\pi}{2} + 4 \right) \text{ units}^3$$

$$\text{or } \frac{3\pi^2}{2} + 4\pi \text{ units}^3$$

Q10.

Let roots be

$$\alpha = k-d$$

$$\beta = k$$

$$\gamma = k+d$$

$$\alpha + \beta + \gamma = 3k = -a$$

$$k = -\frac{a}{3}$$

At POI, $y'' = 0$

$$\begin{cases} y' = 3x^2 + 2ax + b \\ y'' = 6x + 2a \end{cases}$$

$$6x + 2a = 0$$

$$6x = -2a$$

$$x = -\frac{a}{3}$$

So the POI occurs when $x = k$, and this is one of the roots so $y\text{-value} = 0$.

$\therefore$ Coord's of POI $(k, 0)$ or $(\beta, 0)$.